

Genetic aspects of seed harvests for the artificial regeneration of wild service tree (*Sorbus torminalis* [L.] Crantz)

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Abstract Isoenzymes were used to study the genetic variation of 121 mature trees of *Sorbus torminalis* in a natural stand and a subset of 21 high yielding seed trees in the same stand used for a commercial harvest. Eight enzyme systems were studied. The genetic variation of the mature stand was higher than that of other stands reported in the literature. However, this level of genetic variation was not represented to the same extent by the seeds of the commercial harvest. Differences between the bud sample taken from the mature stand and the seed sample taken from the commercial subset were particularly obvious with respect to diversity measures and heterozygosities. The hypothetical gametic multilocus diversity (HGMD) of a bud sample of the mature *Sorbus torminalis* trees was more than 70% higher (94.1) than the value of the seed harvest sample (55.1). Deviations between the genetic structures of both samples were statistically significant. There is no indication that inbreeding is an essential component of the mating system of *Sorbus torminalis*. The potential causes for the lack of *Sorbus torminalis* seedlings and saplings originated from seeds and their consequences for regenerating this species are discussed. This research underscores that sampling a forest tree population for commercial harvest can result in a statistically significant reduction in the genetic diversity of the sampled group relative to the overall population.

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Zusammenfassung Anhand von acht Enzymsystemen wurde untersucht, inwieweit sich die genetische Variation der 121 Elsbeeren eines 3,7 ha großen ehemaligen Mittelwaldbestandes mit jener von Elsbeerensamen deckt, die im Rahmen einer kommerziellen Ernte der Früchte von 21 ausgewählten Altbäumen gewonnen wurden.

Die genetische Variation des Altbestandes erwies sich, verglichen mit den Werten anderer in der Literatur beschriebener Populationen, als hoch, spiegelte sich in der genetischen Variation der Samen einer Stichprobe von 21 Altbäumen allerdings nur eingeschränkt wieder. Dies gilt vor allem hinsichtlich der Diversitätsmaße und der Heterozygotie. So lag beispielsweise die hypothetische gametische Multilocusdiversität (HGMD) des Altbestandes um über 70% (94.1) über dem Vergleichswert der Samenstichprobe (55.1). Die Abweichungen zwischen den genetischen Strukturen beider Stichproben sind statistisch signifikant. Es gibt keine Anzeichen, dass Inzucht eine prägende Komponente des Paarungssystems von *Sorbus torminalis* ist. Die Bedeutung der Ergebnisse für die Bemühungen um eine erfolgreiche Verjüngung der selten geworden Baumart *Sorbus torminalis* werden diskutiert.

Keywords Natural regeneration · Mixed stand · Isoenzymes · Genetic variation · Silviculture · Sampling · Population genetics

Introduction

Wild service tree (synonymous: Chequer tree, *Sorbus torminalis* [L.] Crantz) recently proved to yield high economic returns (Drapier 1993a; Wilhelm and Ducos 1996; Kleinschmit 1998) due to its highly valuable wood which is mainly used for the production of decorative furniture (Kausch-Blecken von Schmeling 1994). However, in the past it has rarely been considered in forest management in Central Europe (Röhrig 1972; Wilhelm 1993; Kausch-Blecken von Schmeling 1993; Kahle 2004). This fact can be explained on one hand by a shift in forest management from the coppice with standard system to the high forest system over the last two centuries (Hasel 1985) and on the other hand by the autecology of *Sorbus torminalis*. According to Helms (1998) high forests are stands of trees, generally of seedling origin, that normally develop a high, closed canopy. In contrast the coppice-with-standard-system permanently results in a free crown of the few trees remaining after periodically repeated heavy cuts. It favours tree species like *Sorbus torminalis* which could regenerate by suckers while the high forest system is likely to favour mainly late successional tree species that are highly competitive (Drescher and Majer 1984; Drapier 1993b). Because of its limited ability to deal with interspecific competition, *Sorbus torminalis* is found to survive in mixed stands within high forests only if it is supported by repeated tendings and thinnings (Drescher and Majer 1984; Wilhelm 1993; Schüte and Beck 1996; Wilhelm and Ducos 1996; Müller et al. 2000). Nevertheless, as Kahle (2004) recently showed by analysing various mixed stands with a high percentage of wild service tree, this species can exist in deep shade for a long period. For substantial growth however, shading by other trees should be avoided. The wild service tree can therefore be classified as a light demanding but shade tolerant species (Kahle 2004).

The second reason for the rareness of *Sorbus torminalis* in high forests is that sexual regeneration is found to be very poor (Rasmussen and Kollmann 2004). Comparable to many tropical species which frequently occur as widely scattered individuals (Hodge et al.

2002), *Sorbus torminalis* often propagates vegetatively. However, the reasons for the low amount of seedlings and saplings of *Sorbus torminalis* originating from seeds are not fully known. Oddou-Muratorio et al. (2001b) recently showed that the reproduction of *Sorbus torminalis* seems to be characterized by extinction—recolonization dynamics which could partially explain its rare natural regeneration. This means that *Sorbus torminalis* tends to disappear from a site during succession because of its rare regeneration but might be able to recolonize the site in later stages. Thus under natural conditions *Sorbus torminalis* probably does not permanently contribute to the species composition of a woodland at a given site. In addition, some stands may show a genetic imbalance which could result in inhibited reproduction by seeds.

Whereas vegetative reproduction appeared particularly limited in a population studied by Oddou-Muratorio et al. (2004) other authors assume that sexual reproduction of wild service tree is generally less important than the regeneration by root suckers (Drapier 1993b; Ewald et al. 1994). In fact natural regeneration regardless whether vegetative or sexual, is missing in many cases. For that reason artificial regeneration by planting is often recommended (Röhrig 1972; Drapier 1993b). For this purpose seed harvests are necessary. However, these harvests often do not consider every individual of a wild service tree population, but restrict the collection of seeds to the very dominant and/or strongly fruiting trees.

Against this background the objectives of the study were (i) to determine whether low genetic variability of the mature trees could serve as an explanation for the very low levels of natural regeneration of the investigated stand. Based on a genetic inventory of the target stand, we (ii) tested whether the genetic variability of the investigated population is represented in the seeds of a commercial harvest.

Materials and methods

Site description

The study was conducted in a stand which was previously investigated by Müller et al. (2000). It is located approximately 55 km northwest of the city of Bamberg and 10 km west of the city of Bad Königshofen im Grabfeld (10°21'41" E, 50°17'11" N) at an altitude of 320 m above sea level (Fig. 1). The area is on a slight easterly incline (5°). The climate is classified as warm and dry with a mean annual temperature of 9°C and a mean annual precipitation of about 600 mm. The dominant soil type is a Typic Eutrucept (US soil taxonomy). It can also be classified as a Eutric Cambisol according to the world reference base (WRB) soil classification. The humus type is mull. The ground vegetation consists mainly of calciphilous species.

Composition and structure of the mature stand

A detailed analysis of the mature stand structure is given by Müller et al. (2000). From this study it can be seen that only 4% (41 out of 1,042 individuals) of the trees were found in the overstory, which was defined as the height class comprising 75–100% of the top tree height. However, these few individuals represent 37% of the stand basal area (23.22 m²) (Müller et al. 2000). While sessile oak (*Quercus peatrea* (Matt.) Liebl.) dominated the overstory (76% of stem number and 78% of basal area), Wild service tree occurred mainly in the co-dominating layer i.e., within the height class ranging from 50 to 75% of the top tree height. In this height class 38% of all trees (39% of the basal area) were wild service trees.

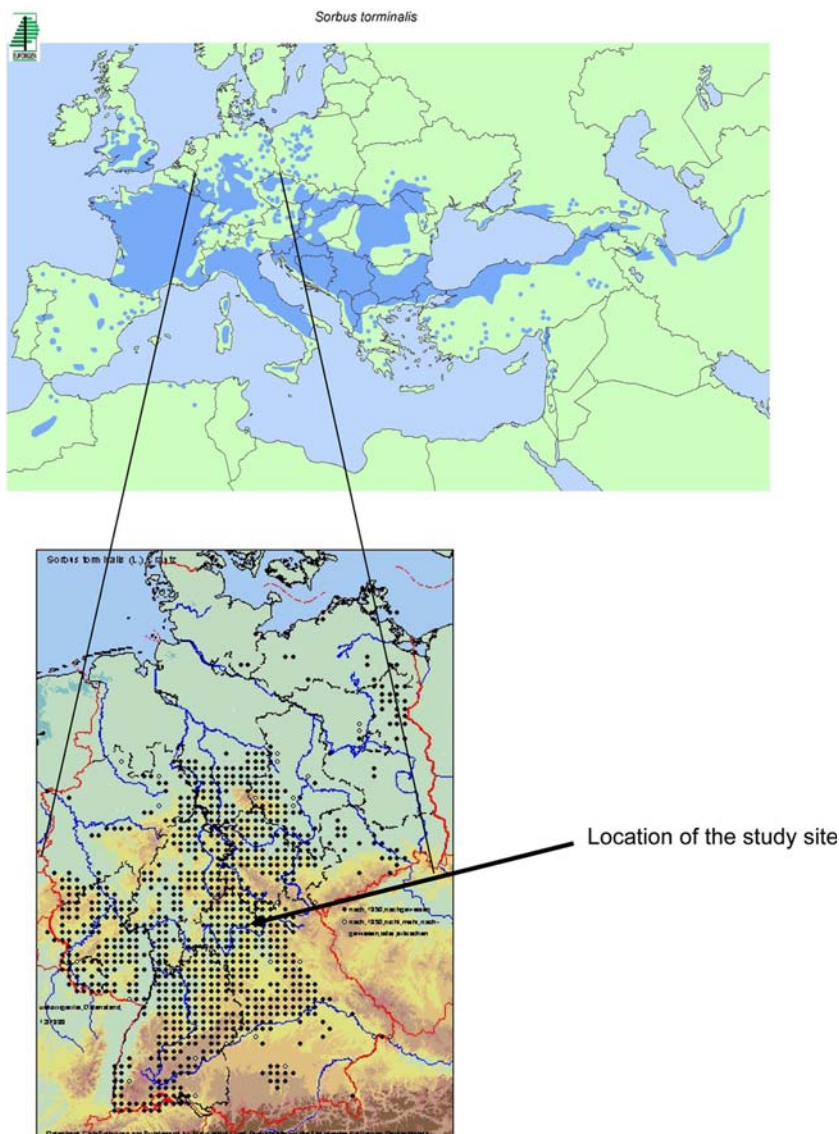


Fig. 1 Distribution range of *Sorbus torminalis* [L.] Crantz in Europe (above, Demesure-Musch and Oddou-Muratorio 2004) and in Germany (below, Bundesamt für Naturschutz 2005)

These trees were not randomly distributed throughout the stand but showed a clumped pattern (Müller et al. 2000). Only a few wild service trees could be found in the overstory (4%) and in the understory (height class 50% maximum tree height) (1%). The understory was dominated by beech (*Fagus sylvatica* L.) and hazelnut (*Corylus avellana* L.).

Genetic variation in mature trees and a bulked seed collection

As was the case in other studies (e.g., Demesure et al. 2000) the genetic variation between the buds of mature trees and seeds from a subsample of these trees was investigated by

utilizing isoenzyme markers. The buds were taken from all 125 mature wild service trees from the studied stand using a shot gun and shooting 1–3 times into the crown of each tree of the mature stand in 1996. Buds from 121 trees could be analysed. In the same stand seeds of the 21 most seed-bearing mature trees were harvested by a commercial nursery in 1994. The number of trees considered for the seed harvest is within the range of common sample sizes of genetic provenance tests (Hodge and Dvorak 2001). One hundred and eighty-seven of these seeds were analysed within this study. Thus the genetic variation of the mature stand (buds from 121 trees) was compared with that of 187 seeds which originated from a commercial harvest of only 21 trees. Genetic control and inheritance of isoenzymes was inferred from other deciduous tree species such as *Fagus sylvatica* (Müller-Starck and Starke 1993) and *Quercus petraea* (Müller-Starck et al. 1996) based on segregation studies among full-sib families. Enzymes from buds and seed tissues were separated by starch-gel electrophoresis (for a detailed description of the methods used see Müller-Starck 1998; Müller-Starck et al. 2001). Thus to genotype the seeds material from the embryos was analysed. Enzyme systems with tissue dependent expression of isoenzymes such as acid phosphatases or esterases were excluded from further studies. Eight polymorphic enzyme systems were studied (abbreviations and EC No in brackets): Aconitase (ACO, 4.2.1.3), Alcohol-Dehydrogenase (ADH, 1.1.1.1), Aspartate-Amino-Transferase (AAT, 2.6.1.1), Isocitrate-Dehydrogenase (IDH, 1.1.1.42), Malate-Dehydrogenase (MDH, 1.1.1.37), Phosphoglucosomutase (PGM, 2.7.5.1), 6 Phosphogluconate-Dehydrogenase (6PGDH, 1.1.1.44), Phosphoglucose-Isomerase (PGI, 5.3.1.9). Genotypes were scored at nine polymorphic gene loci: ACO-A, ADH-B, AAT-C, IDH-A, MDH-A, PGM-C, 6PGDH-B, PGI-C, PGI-D.

Intrapopulational variation was measured by means of the observed and the conditional heterozygosities (H_a , H_c , Gregorius et al. 1986). The number of alleles and genotypes per locus, the number of polymorphic gene loci, the allelic diversity, the multi-locus diversity and the population differentiation δ_T (Gregorius 1987), the hypothetical gametic multilocus diversity v_{gam} (Gregorius et al. 1985) and the potential genetic diversity M_{Geno} (Gregorius et al. 1979) were also determined. Interpopulational variation was quantified by genetic distance (Gregorius 1974), population differentiation (Gregorius and Roberds 1986), the χ^2 -homogeneity test and the G -test. In addition to the Hardy–Weinberg structure, the fixation coefficient F (Wright 1969) was calculated.

Results

Genetic variation in the mature stand and the seed sample

A total of nine putative gene loci were polymorphic (Table 1; further details see Biedenkopf 1999). The locus AAT-C was weakly expressed in seed tissues with correspondingly faint staining of bands. Consequently, this locus had to be omitted in the genotyping of seeds.

Intrapopulational variation in buds and seeds samples

Bud samples from 121 trees and a total of 187 seeds originating from a subset of 21 fruiting trees (pooled sample) were genotyped. In Table 1, variation parameters are surveyed. The average number of alleles per locus was the same in the bud and the seed sample ($A_L=2.50$). Also the average number of genotypes per locus was identical in both samples ($G_L=3.75$).

Table 1 Observed number of alleles and genotypes per locus (A_L , G_L), genic (allelic) population differentiation δ_T , actual (observed) [H_a] and conditional heterozygosities [H_c], and corresponding mean values (gene pool values in case of δ_T) in bud and seed samples

Gene locus	Buds					Seeds				
	121					187				
	A_L	G_L	δ_T	H_a (%)	H_c (%)	A_L	G_L	δ_T	H_a (%)	H_c (%)
ACO-A	2	3	0.502	53.7	53.7	2	3	0.499	33.2	35.6
ADH-B	3	4	0.499	47.1	53.3	3	5	0.469	44.8	65.1
IDH-A	2	2	0.025	2.5	100	2	2	0.016	1.6	100
MDH-A	2	3	0.488	57.0	68.3	2	3	0.501	54.5	56.7
6PGD-B	3	4	0.313	31.4	86.4	3	3	0.057	5.9	100
PGI-C	3	6	0.588	62.8	62.8	3	5	0.510	36.4	41.5
PGI-D	3	5	0.420	46.3	88.9	3	6	0.409	32.1	63.8
PGM-C	2	3	0.494	41.3	47.2	2	3	0.500	44.9	47.2
Total	20	30				20	30			
Mean	2.50	3.75	0.416	42.8	70.1	2.50	3.75	0.370	31.7	63.7

In the case of the population differentiation, δ_T , which is 0.0 in case of fixation and approaching 1.0 with increasing number of evenly distributed genetic types, in both samples 75% of the values were greater than 0.40 (see Table 1). The highest values were observed at the locus PGI-C (0.588 vs. 0.510). The smallest values were found at IDH-A (0.025 vs. 0.016). Pronounced differences were observed at the gene locus 6PGDH-B (0.313 vs. 0.057) and PGI-C (0.588 vs. 0.510). The multi-locus mean value of the bud sample was 0.416 and that of the seed sample, 0.370.

The observed heterozygosities (H_a) of the bud sample ranged between 2.5 in the case of IDH-A and 62.8 in case of PGI-C. The mean was 42.8. The corresponding mean value of the seed sample was 31.7, ranging between 1.6 (IDH-A) and 54.5 (MDH-A). In the case of the conditional heterozygosities (H_c), the mean values of the buds and the seeds were 70.1 and 63.7, respectively, ranging from 50 to 100 (see Table 1). In both samples, IDH-A showed maximum heterozygosity (100%). The same value was evident for 6PGDH-B in seeds. The smallest values were 47.2 (PGM-C) in the buds and 35.6 (ACO-A) in the seeds. The mean value of the bud sample was 70.07% vs. 63.73% in the seed sample.

As indicated in Table 2, the allelic diversity in the bud sample varied between 1.025 (IDH-A) and 2.417 (PGI-C) as compared to 1.016 (IDH-A) and 2.033 (PGI-C) in the seed sample. The hypothetical gametic multilocus diversity (v_{gam}) was equal to the number of genetically different gametic types which hypothetically can be produced by the

Table 2 Genic (allelic) diversity (v) and hypothetical gametic multi-locus diversity (v_{gam}) in bud and seed samples

Gene locus	Genic (allelic) diversity v	
	Buds	Seeds
ACO-A	2.000	1.990
ADH-B	1.988	1.877
IDH-A	1.025	1.016
MDH-A	1.947	1.997
6PGDH-B	1.452	1.061
PGI-C	2.412	2.033
PGI-D	1.720	1.689
PGM-C	1.970	1.995
v_{gam}	94.127	55.117

individuals studied based on their eight-locus genotypes, i.e., a measure of genetic variability and thus an indicator for adaptive ability (Gregorius et al. 1986). For the bud sample the value of v_{gam} was more than 70% higher (94.127) than the value of seed sample (55.117).

With the exception of the loci ACO-A, PGI-C, and PGI-D in the case of the offspring sample, and PGM-C in both samples, the F -values were negative or close to zero (see Table 3). The observed genotypic structures did not reveal a general trend of an excess of homozygotes compared with the Hardy–Weinberg proportions. Furthermore, the heterogeneity among the F -values of the eight loci was substantial among the offspring sample. These findings indicate the absence of inbreeding because preferential mating among relatives of the adult population would result in a continuous surplus of homozygotes over the studied polymorphic gene loci. On average, the F -values are greater among the offspring samples compared with the adults, but the excess at a few gene loci is interpreted as the presence of selective effects in favour of homozygotes at these particular loci.

Interpopulational variation in buds and seeds

The results of the testing of frequency distributions (see Table 4) revealed several statistically significant deviations between mature trees (bud sample) and the offspring (seed sample). In the case of the allelic structures, these deviations were indicated by loci ADH-B, MDH-A, 6PGDH-B and PGI-C. In the case of the genotypic structures, the loci ACO-A, ADH-B, MDH-A, 6PGDH-B, PGI-C and PGI-D revealed statistically significant deviations.

The values for the allelic distances were lower than 0.1 except for the gene loci ADH-B, MDH-A and 6PGDH-B. The greatest genetic distance between the samples was indicated by the locus 6PGDH-B (0.280 for the genotypic structure), and the smallest genetic distance was evident in case of IDH-A (0.004 for the allelic structure). The value of the gene pool distance between both samples was relatively small (0.069). As a consequence of the greater number of genotypes as compared to alleles, and the correspondingly lower representation of genotypes in frequency classes, the genotypic distances were greater than the allelic ones. In general, the results therefore indicate that the difference between the genetic structures of the parental and the offspring sample are substantial. Consequently the seed collection cannot be considered as representative of the parental trees in the high forest that was studied.

Table 3 Fixation coefficient F and χ^2 test of deviations between observed genotypic structures and corresponding Hardy–Weinberg proportions

Gene loci	Buds			Seeds		
	χ^2 value	Significance level	F	χ^2 value	Significance level	F
ACO-A	0.669	n.s	−0.074	20.281	***	0.334
ADH-B	1.142	n.s	0.052	2.552	n.s	0.039
IDH-A	0.019	n.s	−0.013	0.012	n.s	−0.008
MDH-A	3.602	n.s	−0.173	1.598	n.s	−0.092
6PGDH-B	3.076	n.s	−0.009	0.172	n.s	−0.028
PGI-C	6.766	n.s	−0.073	22.047	***	0.284
PGI-D	3.721	n.s	−0.106	22.374	***	0.213
PGM-C	3.123	n.s	0.161	1.852	n.s	0.100

Significance levels: $P \leq 0.0001$ (***), n.s.=not significant

Table 4 Statistical comparison between allele frequencies and genotype frequencies of bud and seed samples based on a log likelihood ratio test and χ^2 -test, and corresponding genetic distances

Gene loci	Log likelihood ratio test alleles		χ^2 -test alleles		Allelic distance	Log likelihood ratio test genotypes		χ^2 -test genotypes		Genotypic distance
ACO-A	0.711	n.s.	0.711	n.s.	0.035	13.219	**	13.197	**	0.206
ADH-B	12.025	**	11.345	**	0.117	14.160	**	12.407	*	0.161
IDH-A	0.285	n.s.	0.292	n.s.	0.004	0.288	n.s.	0.295	n.s.	0.009
MDH-A	6.068	*	6.049	*	0.101	7.419	*	7.192	*	0.114
6PGD-B	44.236	***	43.273	***	0.152	45.646	***	43.572	***	0.280
PGI-C	21.980	***	22.114	***	0.085	33.346	***	32.488	***	0.273
PGI-D	0.530	n.s.	0.533	n.s.	0.019	18.445	**	15.205	**	0.142
PGM-C	0.851	n.s.	0.850	n.s.	0.038	1.047	n.s.	1.053	n.s.	0.056

Significance levels: $P \leq 0.05$ (*), $P \leq 0.01$ (**), $P \leq 0.001$ (***), n.s.=not significant

Discussion

A regeneration inventory using 825 sample units of 0.5 m² in size revealed that the number of suckers, seedlings and saplings of *Sorbus torminalis* found in the stand that was sampled was very low (828 individuals ha⁻¹ ($577 < \mu < 1,151$)). This finding indicates that it is not only small stands with a few adult wild service trees, or stands at the distribution limit of this species, which often show inbreeding (Rasmussen and Kollmann 2004), are characterized by poor regeneration. This result therefore confirms other studies that reported very poor reproduction of *Sorbus torminalis* (Röhrig 1972; Kausch Blecken von Schmeling 1981). However, in our case outbreeding depression and hence inhibited regeneration because of genetic imbalances is not likely. This assumption is firstly supported by the high number of individuals in the population. Secondly several other (small) populations can be found within a radius of 50 km, which is important as *Sorbus torminalis* is known as a species with high dispersal abilities (Oddou-Muratorio et al. 2004). Thirdly “old rare” species as defined by Rasmussen and Kollmann (2004) may tolerate low population density and isolation, e.g., by the ability of vegetative reproduction (Rasmussen and Kollmann 2004). This pattern is known for tropical tree species as well (Hodge et al. 2002). However, low population density may counterbalance the high dispersal ability of *Sorbus torminalis* (Oddou-Muratorio et al. 2004). Thus there are concerns about the adaptability of forest tree populations if population densities drop below species-specific threshold densities (Finkeldey and Ziehe 2004). However, the genetic variation of the investigated mature stand (Tables 1, 2, 3 column “buds”) is higher than that of most of the stands in France reported by Prat and Daniel (1993), Demesure et al. (2000), Oddou-Muratorio et al. (2001a, b) and a population in Switzerland (Menn 1998).

As genetic imbalances of the mature stand could not be observed, other factors might be responsible for the low number of *Sorbus torminalis* within the regeneration. However, neither the overstory canopy density nor the cover with ground vegetation and competing seedlings of other tree species could serve as an explanation (data not shown). Browsing by roe deer could have played a role despite the fact that the whole stand of 3.7 ha was fenced. Nevertheless, the fence was not intact in some areas and some browsed seedlings were detected. In general *Sorbus torminalis* is known to be heavily browsed (Röhrig 1972; Drapier 1993a; Ewald et al. 1994). Browsing therefore may explain its rareness in regen-

erations in many cases. Moreover, the impact of selective browsing could be misleading in classifying *Sorbus torminalis* seedlings and saplings as less competitive compared to those of other tree species. In fact the impact of ungulates is often superimposed on the competitive interactions within regenerations (Ammer 1996). It is therefore necessary to distinguish between the real and the apparent competitive ability of a species (Connell 1990). The role of birds in the successful germination of *Sorbus torminalis* seeds is not fully understood. Many studies have indicated that the effect of ingestion of fruits by vertebrate frugivores on seed germination is very important (Traveset 1998; Traveset et al. 2001; Verdú and Traveset 2004). In fact, Paulsen and Högestedt (2002) showed that the passage of *Sorbus aucuparia* L. seeds through bird guts increases germination rate and seedling growth. The results of this study, dealing with a tree species of the same genus as *Sorbus torminalis*, indicate the potential role of avian seed ingestion for the reproduction of Wild service tree. In addition, birds seem to be the main factor for the observed homogenization of the genetic structure of 67 *Sorbus torminalis* stands throughout France which was reported by Demesure et al. (2000). The impact of vertebrate frugivores may also explain the spatial aggregation of even-aged individuals found by Oddou-Muratorio et al. (2004). However, there are many factors which impact germination and growth of seedlings after the seeds have passed through the digestive tract of frugivores such as retention time in the gut, seed size, seed source, and frugivore species (Traveset et al. 2001). At present it is not clear which interactions of these factors enhance the regeneration of *Sorbus torminalis* by seeds and how the assumed positive effect of avian frugivores on *Sorbus torminalis* seeds could be used for the management of regeneration stands containing a considerable number of mature wild service trees.

Because of the variety of problems inherent to the natural regeneration of *Sorbus torminalis*, planting seedlings is a common silvicultural measure. However, there is some evidence that many of the artificially regenerated *Sorbus torminalis* exhibit very limited genetic variation and/or are introduced from regions with other growing conditions (Jacobée 1993). We recommend to harvest at least *Sorbus torminalis* only in stands with a high number of parent trees (Tables 1, 2, 3, 4). Moreover, seeds of almost all trees should be harvested. These measures might help to avoid increased expressions of recessively deleterious alleles as a consequence of preferential mating among relatives. Deficiencies in the genetic variation of the offspring sample mainly reflect diversity measures and heterozygosities. Lower genetic diversities in the offspring sample, in spite of similar or identical numbers of genetic types in both samples are a consequence of differences in the frequency distributions which tend to be more even in the parental distributions compared with the offspring (see Table 1). The applied diversity measures increase with increasing evenness of the frequency distribution and are identical to the number of genetic types in the case of equal representation of genetic types. The surplus of 70% of the hypothetical gametic multilocus diversity of the adult sample compared with the offspring sample indicates the relative unevenness of the latter (see Table 2). This trend is also evident in the case of the population differentiation, δ_T . Consequently, the risk of losing genotypes and alleles is greater in the offspring sample compared with the adult sample. A similar trend is evident in case of the heterozygosities which are lower among the seed sample compared with the offspring sample. Generally, the observed large proportions of heterozygotes in *Sorbus torminalis* compared with other tree species (Müller-Starck et al. 1992), support the protection of genetic types which are too rare to occur as homozygotes. In this sense, the seed sample furthermore is subjected to a greater risk of losing rare information than the adult sample. This result causes concern as adaptability of forest resources, which includes genetic variance, is a prime objective of forest management (Lefèvre 2004). Thus

the consequences of an inadequate size of seed samples might result in genetic effects comparable to those caused by forest fragmentation and deforestation (Rossetto et al. 2004; Baucom et al. 2005).

An important finding is, that in spite of the rare occurrence of *Sorbus torminalis*, inbreeding depression does not appear to be an essential component of the mating system of this species. The present study does not reveal evidence for genetic loads following inbreeding among the seed sample (see Table 3). Finally, the statistical comparison between the genetic structures among the offspring and the seed sample demonstrates that the seed collection cannot be regarded as being representative of the studied adult sample. Commercial seed harvests should therefore comprise as many parent trees as possible. However, these results warrant confirmation by additional investigations using other stands and considering different years of fructification.

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